

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- No Cummins® Service Tools required.

Additional Service Items

- Loctite® 242, Part Number 3824040.

General Information

- All options utilize a Cummins® supplied molded hose elbow at the air compressor intake and connect to a Cummins® supplied air inlet tube, mounted to the crankcase breather bracket.
- After the air compressor inlet tube, there are two routing options to the connection point in which air is drawn.

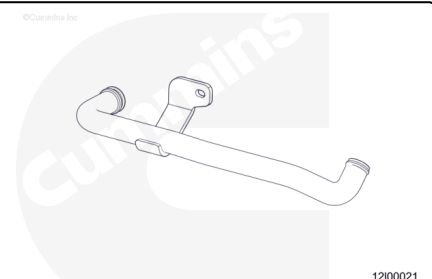


Note : There is a resonator tube option that may impact air the compressor inlet tube routing on a limited number of chassis.

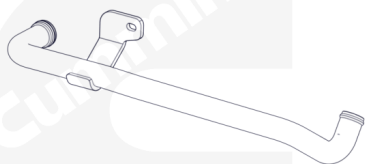
1. Molded elbow that connects the head of the air compressor to the cold side air compressor inlet tube.

2. Cold side metal tube assembly mounted to the breather bracket that connects the molded elbow to the front tube.

For a single cylinder air compressor inlet tube, the tube should be shorter.

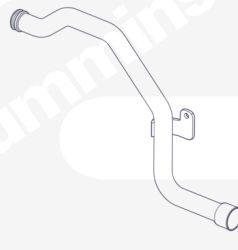


For twin cylinder air compressor inlet tube, the tube should be longer.



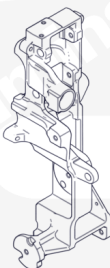
12100022

3. Front metal tube connecting the alternator bracket casting to the cold side metal tube assembly.



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4. Alternator bracket casting internal air passage.



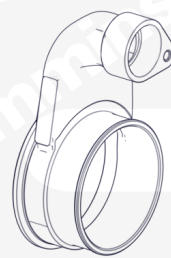
12400082

5. Hot side air compressor inlet tube connecting the turbocharger adapter to alternator bracket tube.



12100024

6. Turbocharger adapter connecting the hot side air compressor inlet tube to the turbocharger compressor inlet.



12400083

Preparatory Steps

⚠ WARNING ⚠

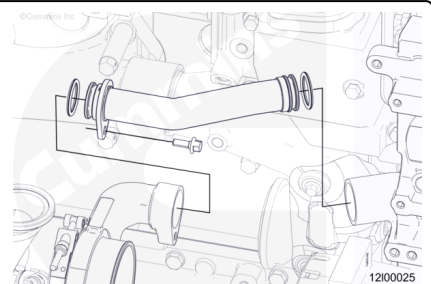
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. See equipment manufacturer service information.

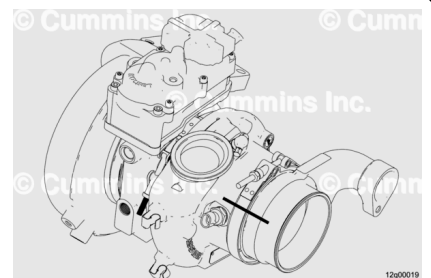
Remove

Remove the capscrew from the turbocharger adapter on the hot side of the engine.

Slide the hot side air compressor inlet tube out of the alternator bracket casting.



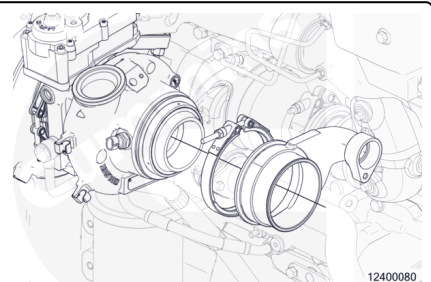
Use a paint pen to mark a single line across the turbocharger compressor housing, v-band clamp, and air compressor adapter. This paint pen mark is to identify the orientation of all three components.



Loosen v-band clamp holding the turbocharger adapter to the turbocharger compressor housing.

Remove turbocharger adapter.

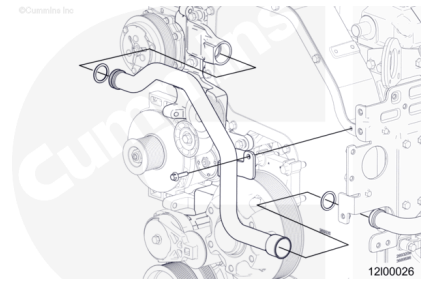
Remove and discard o-ring from the adapter to turbocharger compressor housing, if equipped.



Remove the capscrew from the front metal tube mount on the gear housing.

Remove the v-band clamp.

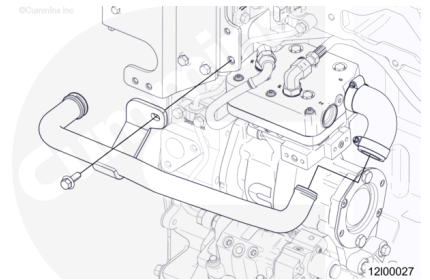
Slide front metal tube out of the cold side metal tube assembly and the alternator bracket casting.



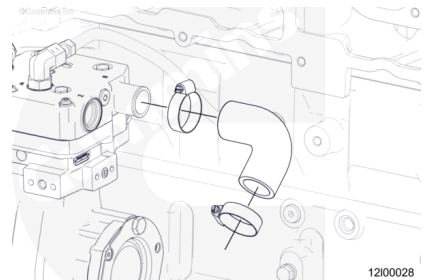
Loosen the v-band clamp on both ends of the molded elbow.

Remove the capscrew from the cold side metal tube assembly mount on the breather bracket.

Slide the cold side metal tube assembly out of the molded elbow.



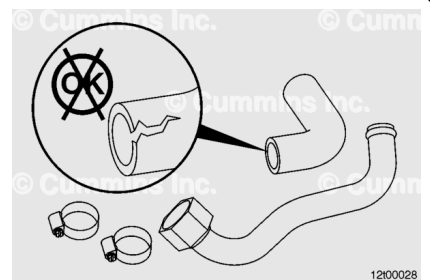
Slide out the molded elbow from the air compressor head.



Inspect for Reuse

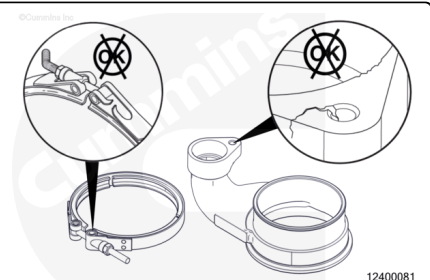
Inspect the air compressor inlet tube for cracks or other damage.

Replace the inlet tube if cracks or other damage is found.



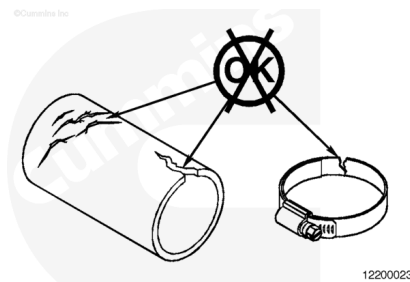
Inspect turbocharger adapter and v-band clamp for cracks or other damage.

Replace turbocharger adapter and v-band clamp if cracks or other damage is found.



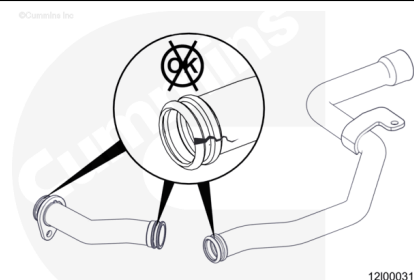
Inspect the hose, v-band clamp and the hose clamps for cracks or other damage.

Replace hose, v-band clamp and hose clamps if cracks or other damage is found.



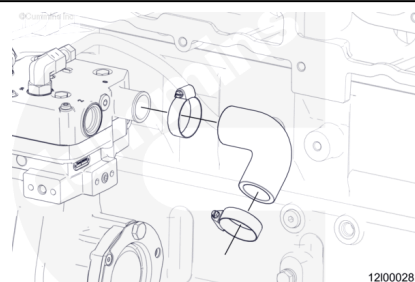
Inspect the o ring mating surfaces of front metal tube and hot side air compressor inlet tube for cracks or other damage.

Replace components if cracks or other damage is found



Install

Install the molded elbow into the air compressor head.

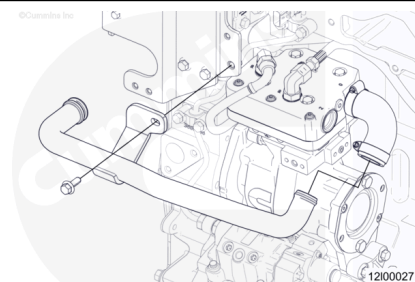


Replace the o-ring on the cold side metal tube assembly.

Install the cold side tube assembly into the molded elbow and tighten the mounting capscrew on the breather bracket.

Torque Value: 47 n•m [35 ft-lb]

Tighten the v-band clamps on both sides of the molded elbow.



Replace the o-ring on the front metal tube.

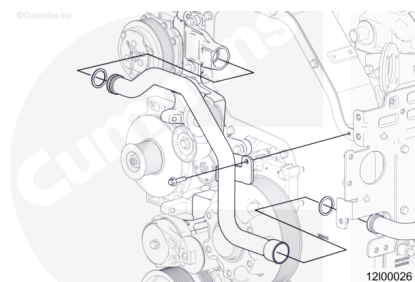
Install the front metal tube into the cold side tube assembly and the alternator bracket casting.

Use a lint free cloth to wipe any debris or loose thread locker residue from capscrew.

Apply Loctite® 242, Part Number 3824040, thread locker to the capscrew threads.

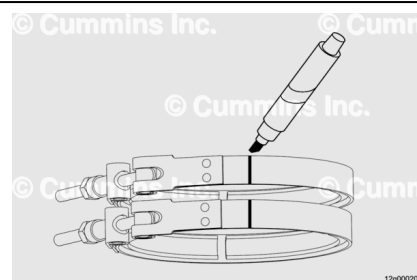
Tighten the mounting capscrew to the gear housing.

Torque Value: 31 n•m [23 ft-lb]

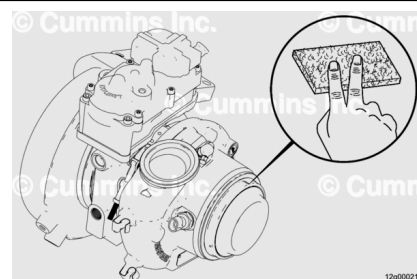


For adapters without an o-ring:

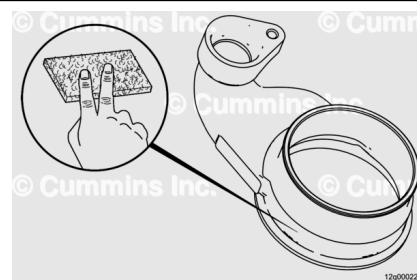
Transfer the markings from the original v-band clamp to the new v-band clamp, if necessary.



Clean the face of the turbocharger compressor housing.

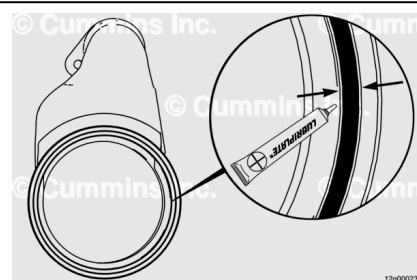


Clean the face of the air compressor adapter.

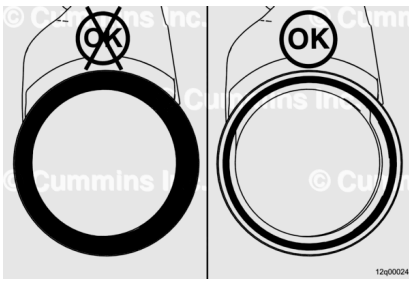


Apply RTV on the flat surface of the outside edge of the turbocharger compressor adapter's rear face.

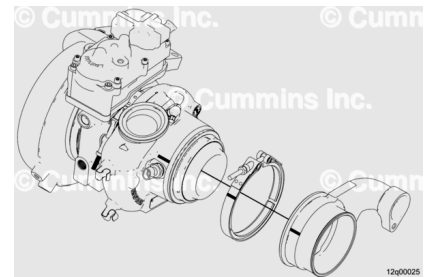
1. Bead of RTV must be no more than 3.175 mm [0.12 in] wide along the outer diameter of the face.
2. Verify the bead or RTV is both even thickness and even height.
3. Verify application of RTV is limited to the outside edge of the illustrated face.



Examples of correct and incorrect RTV application shown in illustration.

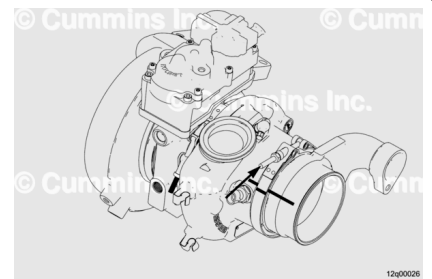


Install the air compressor adapter and v-band Clamp within 1 minute of RTV application to the face of the turbocharger compressor assuring proper alignment of the paint pen to mark a single line across the turbocharger compressor housing, v-band clamp, and air compressor adapter. This paint pen mark is to identify the relative local orientation of all three components orientation.



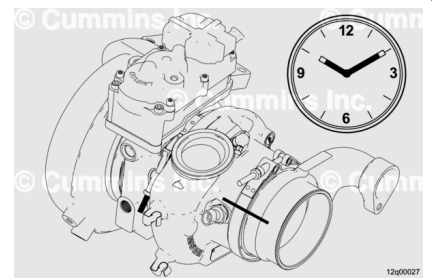
Torque the v-band clamp securing the components back together keeping the clocking marks of all three components aligned.

Torque Value: 12.5 n·m [111 in-lb]



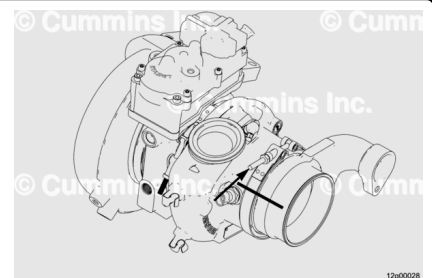
Allow the RTV to cure for four hours before starting the engine to minimize risk of drawing RTV into the turbocharger.

Note : If vehicle needs to be relocated, engine can be started if operation time is limited to less than 5 minutes and vehicle speed does not exceed 15 mph.



After allowing the RTV to cure for at least four hours, verify the torque on the v-band clamp.

Torque Value: 12.5 n·m [111 in-lb]



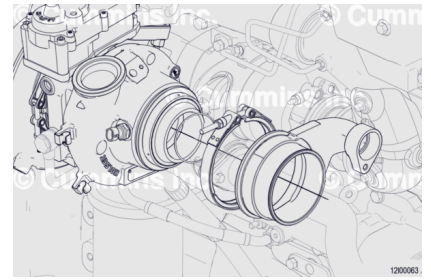
For adapters with an o-ring:

Install a new o-ring on the adapter to turbocharger assembly.

Install the turbocharger adapter onto the turbocharger compressor housing with v-band clamp.

Tighten the v-band clamp.

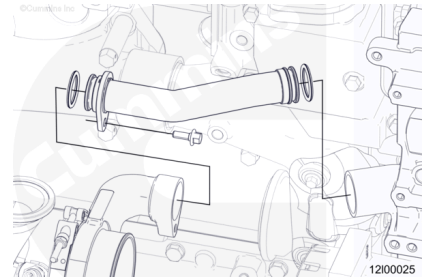
Torque Value: 12.5 n•m [111 in-lb]



Install the hot side air inlet tube to the alternator bracket casting and the turbocharger adapter.

Tighten the capscrew to the turbocharger adapter.

Torque Value: 20 n•m [177 in-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Connect the batteries. See equipment manufacturer service information.
- Operate the engine and check for leaks and proper operation